

Single-Camera Effective Parameter Identification for Bimanual Dough Simulation

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Abstract—Learning bimanual manipulation of deformable materials requires models that connect camera observations, tool motion, and material response. We present TaichiDough, a real-to-simulation pipeline for dough observed by one calibrated RGB-D camera while two spatulas follow recorded trajectories. A metric camera–scene calibration and floor-aware reconstruction initialize a mass-preserving three-dimensional Moving Least Squares Material Point Method model. The corrected Taichi simulator replays signed-distance-field tools and differentiates a partial-view depth objective through the full particle trajectory. Material variables are optimized in scaled latent coordinates, while an independent full-resolution evaluator measures visible-geometry prediction. On one recorded sequence, a five-parameter run reduced its training objective by 61.8% before interruption. A separate completed run improved a strict temporal-continuation score over frozen geometry, but reached the Young’s-modulus bound and used mismatch-tolerant CUDA replay. We therefore interpret the estimates as configuration-dependent effective parameters rather than uniquely identified dough constants. The results establish the sensing, simulation, and optimization basis for future synthetic-data generation and robot-controlled bimanual shaping, while making the present identifiability and numerical limitations explicit.

I. INTRODUCTION

Robotic manipulation of dough and other elasto-plastic materials is difficult because an action changes both the visible geometry and the latent mechanical state. This complicates bimanual shaping with tools: a controller must reason about large deformation, contact, rate dependence, and deformation that does not recover after unloading. Data-driven approaches can learn such behaviour from demonstrations or interaction data, but collecting broad physical datasets with robots is expensive. A calibrated simulator could instead generate controlled synthetic trajectories and support model-based action selection, provided that its predictions are tied to measurements of the real material.

This work develops TaichiDough, a real-to-sim pipeline intended for future bimanual dough shaping with two robot-held spatulas. A calibrated RGB-D camera records the dough while motion capture provides the poses of both tools. The initial visible shape is reconstructed into a volumetric particle model, measured tool motion is replayed in a three-dimensional Material Point Method (MPM) simulator, and material parameters are adjusted to reduce disagreement between rendered and recorded depth observations. The same forward model supports a derivative-free Young’s-modulus

search and checkpointed reverse-mode differentiation through complete trajectories.

The central question is deliberately narrower than camera-based rheometry: what *effective simulator parameters* can one calibrated view and known tool motion constrain? Absolute camera scale does not remove ambiguity among hidden volume, mass, contact, constitutive response, and numerical discretization. We therefore report parameter estimates only for the declared model and setup, and separate training loss from a frozen full-resolution evaluation. The robot execution and synthetic-data learning experiments are future work; the present study establishes and examines the identification pipeline that they require.

The contributions are:

- a metric single-camera pipeline that combines floor-aware volumetric initialization with synchronized replay of two measured spatula trajectories;
- a differentiable MLS-MPM formulation for partial depth observations, including bounded optimizer coordinates for elastic, viscous, and principal-stretch parameters; and
- a case study using derivative-free and differentiable fitting evidence from recorded dough, including the numerical and identifiability limits that prevent interpreting the current estimates as universal material constants. The runs are not a matched optimizer comparison.

Completed identification pipeline

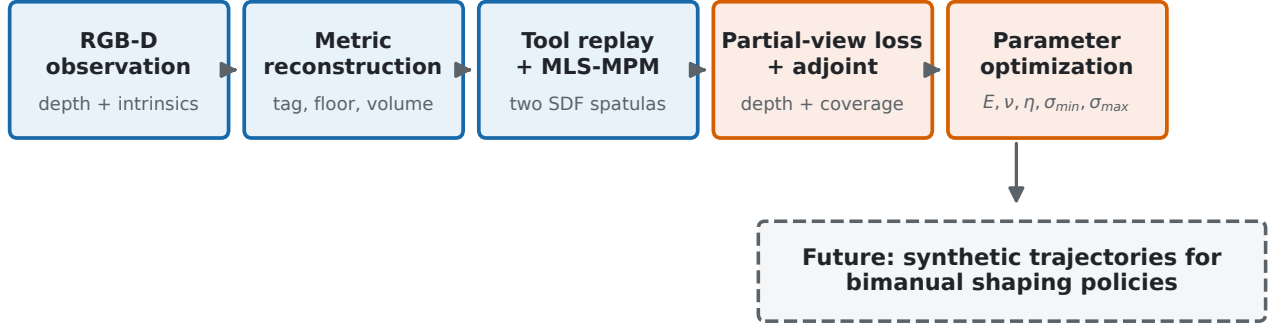


Fig. 1. TaichiDough pipeline. A single calibrated RGB-D camera and recorded poses of two spatulas provide the initial geometry, observations, and boundary motion. A corrected MLS-MPM rollout is projected into the camera and compared with measured depth. Robot execution and synthetic-data learning are intended downstream uses rather than experiments reported here.

II. RELATED WORK

Deformable simulation and differentiable physics. MPM combines Lagrangian material points with an Eulerian grid and is well suited to large deformation and changing topology [1]. APIC and MLS-MPM improve particle-grid transfer and provide practical coupling to rigid bodies [2], [3]; singular-value projection has been used to model elasto-plastic yield in graphics materials [4]. Taichi provides the data-oriented programming system used here [5]. Differentiable simulators such as ChainQueen and DiffTaichi propagate trajectory gradients for control and parameter optimization [6], [7], while PlasticineLab applies differentiable physics to soft-body manipulation [8]. Our contribution is not a new MPM discretization; it connects a corrected MLS-MPM implementation to calibrated partial observations and measured dual-tool motion.

Visual real-to-sim identification. Prior work estimates deformable-object parameters from images, depth, or point clouds. DiffCloud differentiates point-cloud agreement, and Differentiable Depth calibrates soft-body simulations from a single depth camera [9], [10]. DPSI studies differentiable physics-based identification for robotic manipulation of elasto-plastic materials [11]; EMPM combines differentiable MPM with multi-view RGB-D observations and interaction [12]. These systems demonstrate that visual inverse simulation is established. TaichiDough instead studies a constrained setting with one calibrated RGB-D viewpoint, known two-tool trajectories, reconstructed hidden volume, and a visible-geometry objective. This setting makes nuisance parameters and practical identifiability central: visually similar trajectories can result from different combinations of material, contact, mass, and reconstruction assumptions [13], [14].

Robotic shaping and dough mechanics. RoboCraft and RoboCook learn predictive models for shaping elasto-plastic objects with diverse tools [15], [16]; DoughNet focuses on visual prediction for topological dough manipulation [17]. These works motivate synthetic data and predictive models for bimanual shaping, but do not remove the need to

connect a simulator to a particular physical preparation. Food-engineering studies identify dough rheology with controlled loading and force-related measurements. Fabbri and Cevoli, for example, invert a finite-element extrusion model and compare the estimates with capillary rheometry [18]. Dough response also depends on formulation, rate, preparation history, and adhesion [19], [20]. Consequently, the Young’s modulus, viscosity, and stretch limits fitted here describe the selected constitutive and contact model over the recorded motion; they are not claimed as complete rheological properties of dough.

III. METHOD

We formulate calibration as a trajectory-conditioned inverse problem. Let z_t denote the simulated particle state, u_t the measured poses and velocities of the two spatulas, θ the fitted material parameters, and ψ the fixed reconstruction, contact, camera, and numerical settings:

$$z_{t+1} = \mathcal{S}_{h,\Delta t}(z_t, u_t; \theta, \psi), \quad y_t = \mathcal{H}_c(z_t) + \delta_t + \epsilon_t. \quad (1)$$

Here $\mathcal{H}_c$ projects particles into the calibrated depth camera, while δ_t and ϵ_t denote model discrepancy and measurement noise. The estimated values are therefore effective parameters of this specified experiment and simulator, rather than universal dough constants.

A. Metric Reconstruction and Tool Replay

A single RGB-D camera records depth and colour, while the motion-capture system records the two spatula frames. Camera intrinsics and a rigid camera-to-scene transform are obtained from repeated AprilTag/mocap correspondences. The calibration also stores the table plane. All point clouds are transformed once into metric scene coordinates; no scale, translation, or rigid registration is fitted during evaluation.

The first observed dough surface is voxelised and each occupied vertical column is filled down to the calibrated table. Given reconstructed volume V , fixed density ρ , and N sampled material points, the dataset uses

$$M = \rho V, \quad V_p = V/N, \quad m_p = M/N. \quad (2)$$

Thus, episode mass is derived from reconstructed volume, and total volume and mass remain unchanged when particle count is varied. Tool positions are interpolated linearly and quaternion orientations by shortest-path spherical interpolation. The UR5e and Gen3 spatulas are represented by signed-distance fields generated from registered meshes. Their velocity at a contact point x is

$$v_{\text{tool}}(x) = v_{\text{lin}} + \omega \times (x - c), \quad (3)$$

where c is the tool centre. The unilateral Coulomb approximation removes inward normal relative velocity and limits tangential correction using the configured friction coefficient; separating motion is left unchanged. A post-advection projection resolves residual floor or tool penetration.

B. Differentiable MLS-MPM

The simulator implements three-dimensional MLS-MPM with APIC transfer [1] in Taichi [5]. Each particle stores position x_p , velocity v_p , deformation gradient F_p , affine velocity matrix C_p , and optional plastic-volume state. The deformation update is

$$F_p^{n+1} = (I + \Delta t C_p) F_p^n. \quad (4)$$

Young's modulus E and Poisson ratio ν define

$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}. \quad (5)$$

With polar factor R , $J = \det(F_p)$, and viscosity η , the implemented Kirchhoff-style stress is

$$\tau = 2\mu(F_p - R)F_p^T + \lambda J(J-1)I + \eta(C_p + C_p^T). \quad (6)$$

The final term is an additive rate stress, not a multi-time-scale rheological model. For quadratic B-spline weight w_{pi} , particle-to-node offset d_{pi} , and grid spacing Δx , particle-to-grid transfer accumulates

$$m_i += w_{pi} m_p, \quad (7)$$

$$m_i v_i += w_{pi} [m_p v_p + (-4\Delta t V_p \Delta x^{-2} \tau + m_p C_p) d_{pi}]. \quad (8)$$

The corrected grid-to-particle update reconstructs the affine velocity using $4\Delta x^{-2} d_{pi}$. Optional plasticity decomposes $F_p = U\Sigma V^T$ and clamps each principal stretch to $[\sigma_{\min}, \sigma_{\max}]$ before reconstruction. These bounds are dimensionless stretch limits, not yield stresses.

C. Partial-View Objective and Optimization

At each selected frame, a differentiable camera-space particle renderer uses compact splats and soft visibility to produce depth, coverage, and visible particle coordinates in the calibrated camera. After normalizing the configured weights by their sum, the per-frame objective is

$$\mathcal{L}_t = 0.5\ell_d + 0.25\ell_c + 0.25\ell_q, \quad (9)$$

where ℓ_d compares simulated and observed depth change relative to the initial frame, ℓ_c penalizes missing positive observed foreground, and ℓ_q is the directed distance from observed points to visible predicted particles. Unknown pixels

are not treated as empty space. Depth and point distances are divided by the declared 0.01 m scales before applying the Huber penalty. This balances objective components; it does not normalize physical material units. Splat support, visibility, and nearest-particle associations are held fixed within each local derivative evaluation.

Reverse-mode differentiation propagates adjoints through position, velocity, affine velocity, deformation, and plastic history. Host checkpoints bound device memory; each segment recomputes forward intermediates and must normally reproduce endpoint states and contact decisions. Physical gradients are mapped to optimizer coordinates: E uses $q_E = \log(E/E_0)$, while ν , η , and the two stretch limits use bounded scaled coordinates. Projected Adam proposes updates within physical bounds, and Armijo backtracking accepts only finite sufficient-decrease steps. This coordinate transformation avoids comparing raw gradients expressed in Pa, Pa s, and dimensionless units.

We also report an existing derivative-free baseline that varies only E on a logarithmic coarse-to-fine grid while holding its own trajectory setup and other configured quantities fixed. Because it uses different fixed settings, it is evidence for the static search procedure rather than a matched optimizer comparison. Candidate selection uses training frames only. After selection, a separate frozen full-resolution evaluator reports depth, silhouette, coverage, and directed point-distance metrics on the training interval and subsequent temporal continuation. Its strict metric is intentionally distinct from Eq. (9); numerical values from the two objectives are not interchangeable.

Finally, the interpretation separates four parameter families: material ($E, \nu, \eta, \sigma_{\min}, \sigma_{\max}$); interface quantities such as tool friction and floor response; sensing/reconstruction quantities including camera pose, hidden volume, mass, and tool registration; and numerical quantities ($N, h, \Delta t$) and SDF resolution. The present estimates are conditional on all fixed quantities in the latter three families.

IV. RESULTS AND DISCUSSION

We report existing Episode 18 evidence in three groups. They use different simulator revisions or fixed settings and are therefore not pooled as repeated trials. Table I records the role and main limitation of each result, while Figs. 2 and 3 show the camera-matched sequence and differentiable optimization history.

A. Metric Initialization

The metric calibration aggregates 384 moving-tag correspondences and fixes the camera, mocap scene, and floor in metres. At the initial frame, 3970 observed points are compared with 25,495 virtually visible points without fitted translation, scale, or post-hoc registration. On a 3 mm top-view grid, the reconstructed footprint obtains IoU 0.8306, extent ratios 1.077 and 1.106, and an area ratio of 1.191. The centroid error is $(-0.98, 1.25)$ mm, while median and 95th-percentile absolute depth errors are 0.148 and 1.155 mm.

Camera-matched Episode 18 observation and simulation

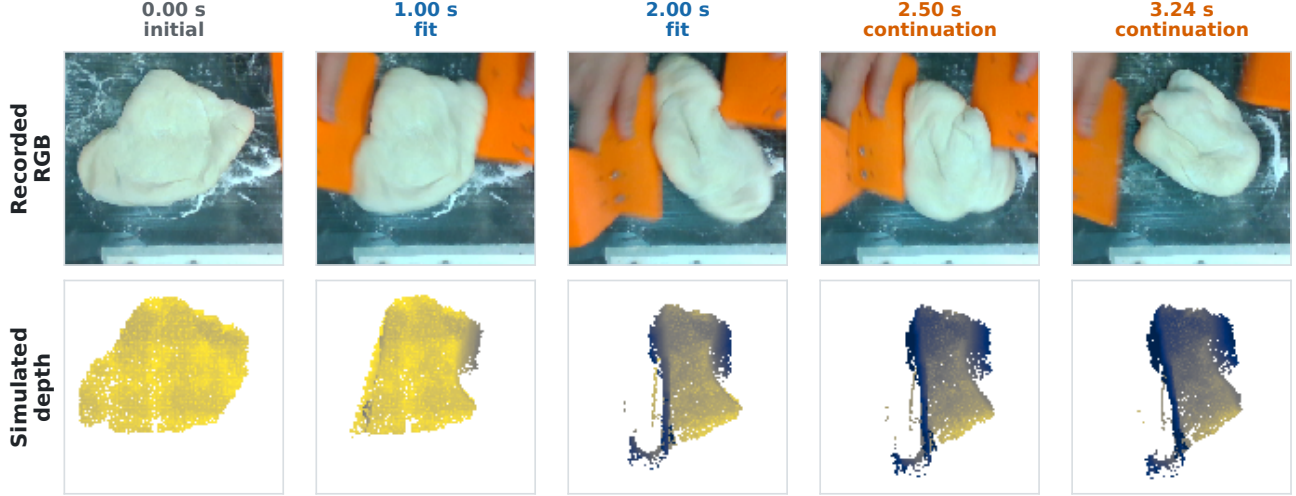


Fig. 2. Exact-timestamp Episode 18 camera views and simulation outputs. The top row is contextual RGB extracted from the original Rosbag; the bottom row is saved dough-only simulated optical depth rendered with the same calibrated camera and display crop. The simulation does not render spatula occlusion. Frames after 2.00 s are temporal continuation of the same motion, not independent generalization.

Interrupted five-parameter fit: accepted-state history

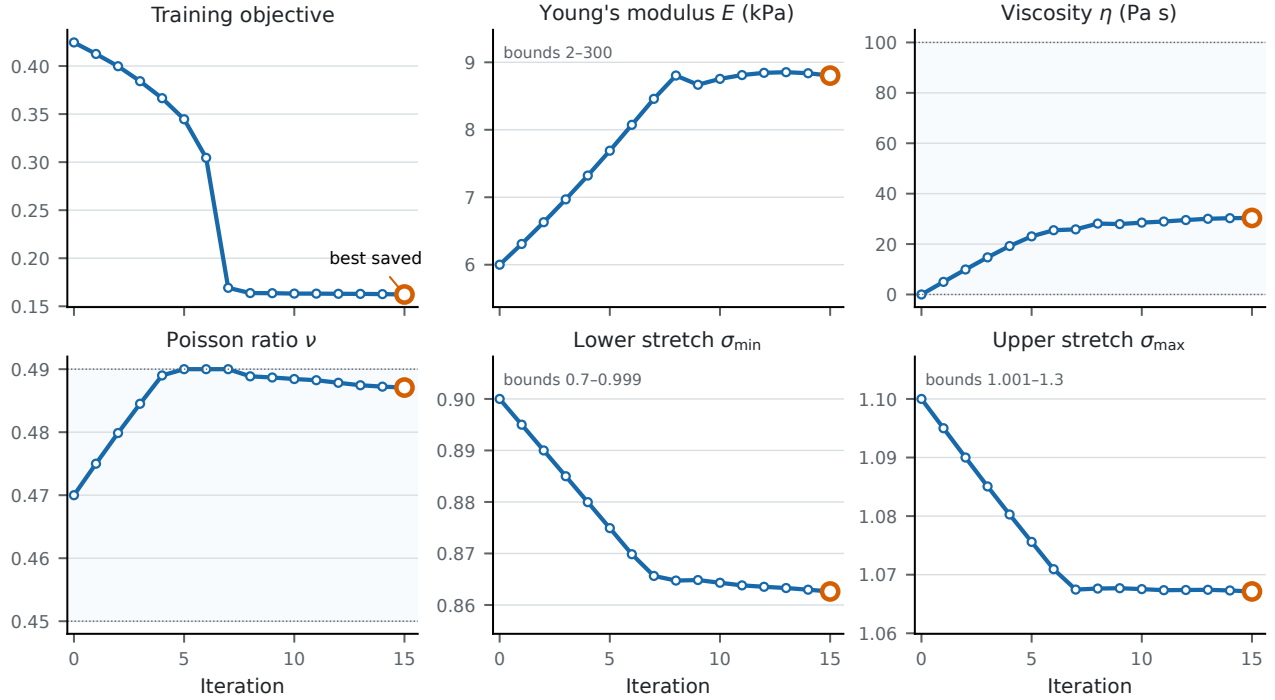


Fig. 3. Accepted objective and parameter history for the interrupted five-variable differentiable run. The orange ring marks the best saved state. Evaluation 18 was interrupted and all completed objectives recorded finite replay mismatches; these are best-so-far effective simulator parameters, not a converged physical estimate.

This supports the initial observation-to-simulation alignment, not the constitutive law.

B. Static and Differentiable Fitting

The derivative-free baseline varies only E with viscosity fixed to zero and plastic projection disabled. Its nominal

candidate is 130.6 kPa, but the result is explicitly unaccepted: candidates from approximately 91 to 187 kPa lie inside the configured near-best tolerance. A bootstrap over the single training window repeatedly selects the same discrete candidate and is not a between-trial uncertainty interval.

The newer differentiable run jointly adjusts E , ν , viscosity

TABLE I

EXISTING SINGLE-SEQUENCE EVIDENCE. THE EXPERIMENTS USE DIFFERENT SIMULATOR REVISIONS AND FIXED SETTINGS AND ARE NOT A MATCHED OPTIMIZER COMPARISON. “CONTINUATION” DENOTES LATER FRAMES OF THE SAME RECORDED MOTION.

Experiment	Result	Interpretation
Metric static initialization	IoU 0.8306; median/p95 depth error 0.148/1.155 mm	Initial visible geometry only; no constitutive validation
Derivative-free E sweep	Best candidate 130.6 kPa; training loss 0.3611	Rejected because the minimum is broad and flat
Differentiable five-variable fit	Objective 0.4246 $\rightarrow$ 0.1623 (61.8%)	Interrupted after 15 accepted updates; mismatch-tolerant replay
Separate strict replay	Train 0.2821 vs. frozen 0.3416; continuation 0.3939 vs. frozen 0.8265	Better visible prediction, but $E = 300$ kPa reaches the upper bound

η , and the two principal-stretch bounds. Seventeen objective evaluations and fifteen accepted updates reduce the partial-view training objective from 0.42455 to 0.16229. The best saved state is

$$\begin{aligned} E &= 8.803 \text{ kPa}, & \nu &= 0.48708, & \eta &= 30.37 \text{ Pa s}, \\ \sigma_{\min} &= 0.86262, & \sigma_{\max} &= 1.06714. \end{aligned} \quad (10)$$

Evaluation 18 ended with a keyboard interruption, so no final selected-parameter record was produced. The saved full-prefix reverse evaluation spans 9839 steps and 59 observations using 155 host checkpoints of length 64. Recomputed segments record 2879 mismatches: 192 endpoint-state arrays and 2687 per-step contact/yield summaries. Maximum endpoint differences are $99.12 \mu\text{m}$ in position, 0.1843 m/s in velocity, 0.02916 s^{-1} in C , and 1.506×10^{-4} in F , although the maximum observation-loss difference is only 3.58×10^{-7} . The small scalar loss discrepancy does not verify the derivative because states and piecewise contact decisions differ. The curve therefore shows that the returned approximate gradient reduced its declared objective; it does not demonstrate convergence, unique recovery, or the exact derivative of the recorded forward trajectory.

C. Frozen Strict Evaluation

A separate completed corrected-physics run exports its selected state to the independent strict evaluator. Its mean four-component score is 0.2821 on frames 1–60, compared with 0.3416 for frozen initial geometry. On frames 61–97 the selected rollout scores 0.3939, compared with 0.8265 for frozen geometry. All expected frames are valid. Mean post-initial pixel IoU is 0.4586 in the fitting interval and 0.4659 over the complete replay report; mean depth-change errors are 4.59 and 5.90 mm for the corresponding training and later-frame scores.

This result shows that deformation under the prescribed replay predicts visible geometry better than leaving the object fixed. It is not a clean validation of identified physical parameters: E reaches its 300 kPa upper bound, viscosity reaches zero, and the run also permits replay mismatches. The later interval uses the same dough, camera, calibration, and recorded manipulation after replay from frame zero. We therefore call it temporal continuation rather than generalization.

D. Limitations for Single-View Identification

A single depth view leaves hidden volume and occluded contact unmeasured, and the evaluator does not render the spatulas into the simulated depth image. Material estimates are consequently conditional on reconstructed volume, mass/density, floor pose, tool registration, contact law and friction, numerical resolution, and loss smoothing. A separate diagnostic also finds an observed table plane about 47.6 mm above the configured floor with approximately 5° tilt, which can be absorbed incorrectly by material or contact parameters. Near-incompressible ν , elasticity, viscosity, and stretch clipping may compensate for one another under a single monotonic trajectory. The present evidence therefore supports an implemented observation-conditioned fitting pipeline and measurable prediction improvement, but not camera-only rheometry or universal dough constants.

V. CONCLUSION

TaichiDough connects a calibrated RGB-D recording and measured two-spatula motion to a mass-preserving MLS-MPM model, partial-view differentiable objective, bounded parameter optimizer, and independent visible-geometry evaluation. Existing Episode 18 results show millimetric centroid/depth alignment with 0.831 footprint IoU, substantial reduction of the differentiable training objective, and better temporal-continuation prediction than frozen geometry. They also show why objective reduction alone is insufficient for physical identification: the derivative-free modulus search is flat, one differentiable result is interrupted, another reaches a parameter bound, and full CUDA replay is not exactly reproducible.

The current values are therefore effective parameters of a stated camera, reconstruction, contact law, constitutive model, and discretization. Future work will use robot-controlled spatulas to collect deliberately informative loading, unloading, hold, and rate-varied motions; repeat measurements across dough preparations; model tool occlusion; qualify full-horizon gradients; and compare matched differentiable and derivative-free optimizers. Once prediction is validated on independent manipulations, the simulator can support synthetic pretraining for inverse models and bimanual dough-shaping policies.

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